Rice variety discrimination by NIR hyperspectral imaging: validation design, accuracy metric and variable selection

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Abstract

Introduction: Near-infrared hyperspectral imaging (NIR-HSI) is widely proposed to authenticate rice varieties, and published accuracies often exceed 95%. We examine how far such figures depend on how the data are divided for validation, on the metric reported, and on band selection.
Methodology: Mean spectra of 500 grains of each of 12 Peruvian varieties were acquired with a Resonon Pika NIR-640 camera (263 bands, 999–1649 nm) and classified with a linear support vector machine on standardized bands (SVM), a linear SVM on unstandardized bands (L-SVM), linear discriminant analysis (LDA) and a neural network (ANN) on the full spectrum and on bands selected by FSCNCA, SFS and CAFS. The classifiers were validated by five-fold cross-validation with randomly assigned grains or whole images held out; band selection was fitted to training data only, and each selected subset was compared against randomly drawn subsets of the same size.
Results: Averaging per-class one-vs-rest accuracies yields $1 - 2e/K$ for any confusion matrix (e being the overall error rate and K the number of classes) and cannot fall below 83.3% for 12 classes; a per-class value of 99.53% for the SVM corresponds to an overall accuracy of 97.2%. With grains divided at random, the SVM and the ANN reached 97% overall accuracy and LDA and the L-SVM 90%; with whole images held out, 88%, 88%, 87% and 84%. Nine-band subsets selected once with all the spectra beat random nine-band subsets clearly only for SFS with the three linear classifiers (by 3.6–7.3 points); FSCNCA led by at most 3.8 points and CAFS was above or below random depending on the classifier; with selection fitted on training data only, SFS beat random subsets with 5–10 bands (by 4–9 points; also with 20 bands for LDA), FSCNCA and CAFS never; likewise, misclassifications concentrated in pairs of spectrally similar varieties.
Conclusions: Reporting overall accuracy, validating images not used for training, and benchmarking band selection against random subsets of the same size would make accuracies comparable between studies. The spectral differences between varieties are consistent with differences in starch, lipid, and water absorption.

Keywords: Food fraud detection, Grouped validation, Rice authentication, NIR hyperspectral imaging, Machine learning, Random-subset baseline, Variable selection

1. Introduction

Rice (Oryza sativa L.) is the staple food of more than half the world's population, providing approximately 20 % of the daily caloric intake of more than 3.5 billion people [1]; it is of particular relevance in Asian countries such as China, India, Thailand, and Indonesia [2]. In China, annual rice production represents approximately 28 % of global production, while in Vietnam, rice constitutes a fundamental pillar of the national economy and food security [3, 4]. Rice composition varies significantly between varieties, with differences in starch structure (amylose and amylopectin content), protein content, lipid content, and volatile compounds that directly influence cooking behavior, texture, aroma, and overall sensory quality [5, 6]. These compositional differences, largely determined by genetic background and growing conditions, explain why some varieties command substantially higher market prices than others.

The growing consumer demand for high-quality rice has generated a market in which premium varieties such as Thai jasmine rice, basmati, Wuyou No.4, and Wuchang aromatic rice command substantially higher prices than conventional varieties [4, 7–9]. This economic disparity has incentivized fraudulent practices involving the adulteration of premium varieties with lower-quality varieties or the falsification of geographical origin, a phenomenon that has reached alarming proportions

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[7, 10]. In the Chinese market, the volume of rice commercialized under the name "Wuchang Daohuaxiang" exceeds the actual production by more than tenfold, evidencing the magnitude of the problem [8, 11]. This situation not only economically harms consumers and legitimate producers, but also compromises trust in quality control systems and food safety [10, 12, 13]. The detection of adulteration in rice grains using economically valuable varieties has therefore become a priority for food regulatory authorities around the world [7, 14].

Traditionally, rice variety identification has been based on morphological methods, chemical analysis, chromatographic techniques (HPLC, GC-MS), molecular assays (PCR, SSR markers), and sensory evaluations [3, 5, 12]. Nevertheless, these methodologies present significant limitations: morphological methods suffer from subjectivity and low precision; analytical techniques require specialized personnel, costly reagents, and considerable analysis time; and molecular methods, although precise, involve laborious and destructive procedures that limit their application in large-scale quality control [3, 13]. Furthermore, the high morphological similarity among numerous cultivated varieties makes their visual discrimination extremely difficult or impossible [4, 15, 16]. These limitations have driven the search for faster, more reliable, and non-destructive analytical approaches capable of addressing the growing demand for rice authentication [12, 17, 18]. Varietal discrimination techniques for the evaluation of adulteration have evolved from destructive chemical methods to non-destructive spectroscopic approaches that preserve sample integrity [19–21].

A rapid review of the state of the art in the evolution of spectroscopic techniques coupled to artificial intelligence tools for rice varietal classification is summarized in Table 1. These studies demonstrate a clear progression from conventional spectroscopy with traditional chemometric methods [4, 16] to hyperspectral imaging (HSI) cameras that simultaneously capture spectral and spatial information, achieving classification accuracies exceeding 98% with VIS-NIR HSI [19]. Similarly, the transition from Machine Learning (ML) to Deep Learning (DL) architectures marked a notable inflection point, CNN demonstrating superior performance for challenging sample matrices, achieving 95.2% accuracy for milled rice using NIR-HSI data [15], while more recent DL architectures, such as CNN-Transformer applied to VIS-NIR hyperspectral data, have attained 95.3% accuracy [22]. This chronological progression underscores the consolidation of spectroscopy and NIR-HSI as the most widely adopted techniques for rice varietal discrimination, with a continuous trend towards more sophisticated instrumentation, and advanced DL architectures.

Despite substantial advances, important challenges remain [12, 18]: a) lack of standardization in data acquisition and pretreatment, b) the absence of public reference databases, and c) model transfer between different equipment and conditions [12, 17, 24]. Moreover, for practical applications, the transfer of models trained under laboratory conditions to real production scenarios remains limited, and different studies have shown a significant performance drop when applied to different years, regions, or equipment [4, 8, 15].

Table 1: State of the art of rice grain varietal classification studies

Year	Varieties	Device	Spectral range	Chemometric technique	Results accuracy (%)	Source
2026	47	Camera	NIR	ML	86.7 - 99.6	[18]
2025	4	Spectrometer	Raman	ML	81.0 - 100.0	[23]
2025	4	Camera	VIS-NIR	ML	75.5 - 98.3	[17]
2025	13	Camera	VIS-NIR	ML, DL	95.3	[22]
2024	5	Camera	Fluor. HSI	ML	95.3	[20]
2024	5	Spectrometer	VIS-NIR, FTIR	ML	78.2 - 100	[9]
2022	3	Camera	NIR	ML, DL	93.0 - 95.2	[15]
2022	4	Spectrometer	NIR	ML	90.0 - 100.0	[16]
2021	4	Spectrometer	FT-NIR	ML, CNN	100	[4]
2020	10	Camera	VIS-NIR	ML	98.6	[19]

Two methodological choices, however, can make reported accuracies hard to compare across studies. The first is the validation unit: grains imaged together share illumination, focus, position and session, so a random split that places grains of the same image in both training and test sets can reward recognition of acquisition conditions rather than of the variety, a form of leakage well documented for data with grouped or hierarchical structure [25, 26]; selecting variables or tuning a model on data that later serve as test data inflates accuracy in the same way [27, 28]. The second is the metric: averaging per-class one-vs-rest accuracies yields values that stay at or above $1-2/K$ even for a classifier that is always wrong, K being the number of classes. In light of these considerations, this study aimed to discriminate twelve Peruvian rice varieties using near-infrared hyperspectral imaging (NIR-HSI) spectra and four classification algorithms: support vector machine (SVM), linear support vector machine (L-SVM), linear discriminant analysis (LDA), and artificial neural network (ANN). Specifically, the study evaluated three variable selection methods (FSCNCA, SFS, and CAFS) against randomly selected wavelength subsets of equivalent size and examined how classification accuracy estimates differ between conventional grain-level random splitting and image-level splitting, in which grains acquired within the same image are assigned exclusively to either the training or test set. Overall accuracy is reported throughout and compared with the mean per-class one-vs-rest accuracy used in earlier studies.

2. Material and methods

Figure 1 illustrates the main steps of the methodology used in this study, including the generation of a database, exploratory analysis, and model building and validation. Image acquisition, grain segmentation, profile extraction and pre-treatment (Section 2.2) were done at UNF with the camera software and MATLAB. Everything downstream of the pre-treated profile matrix – exploratory analysis, classifiers, validation, band selection and statistics – was done in Python 3.14 with NumPy, SciPy and scikit-learn 1.9; the code and the data matrix are available in a public repository (see the Declaration at the end).

2.1. Grains of rice

Half a kilogram of each of the twelve varieties was obtained from the National Institute of Agricultural Innovation (INIA) and transported to the Food Safety Research Laboratory of the National University of Frontera (UNF); and were:

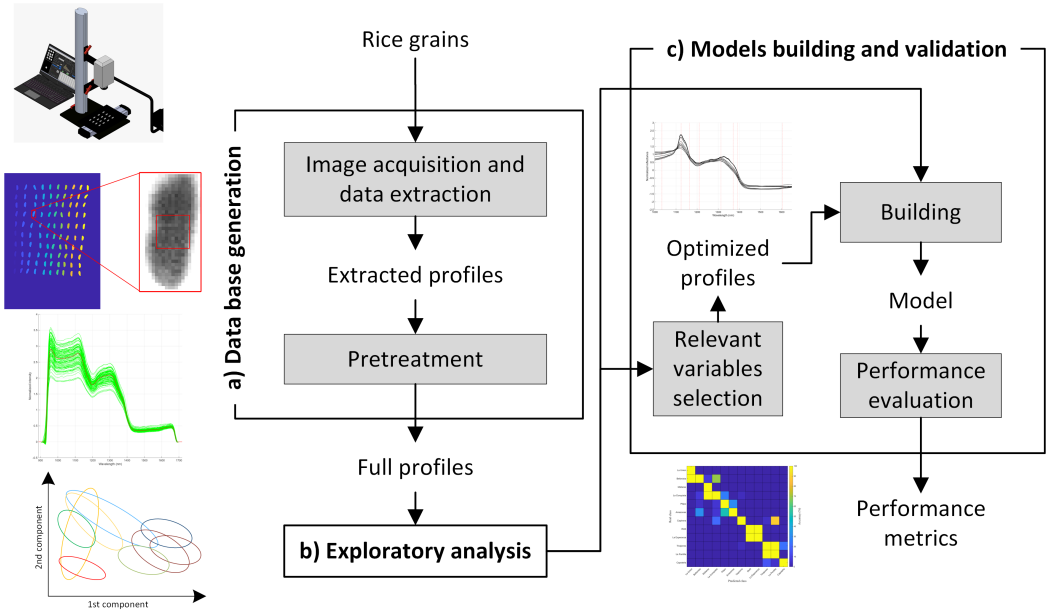


Figure 1: Workflow of the study

- La Unión
- Bellavista
- Mallares
- La Conquista
- Pitipo
- Amazonas
- Capirona
- IR43
- La Esperanza
- Tinajones
- La Puntilla
- Capoteña

The samples were stored in hermetically sealed dark bags under controlled environmental conditions (24 °C, dry environment) to maintain their spectral integrity until image acquisition.

2.2. Database generation

2.2.1. Image acquisition and data extraction

Hyperspectral data were acquired using a Resonon Pika NIR-640 camera (Resonon Inc., USA) in reflectance mode (336 bands, 888–1718 nm; the analysis uses the 263 contiguous bands 47–309, 999–1649 nm); 500 grains of each rice variety were placed on a dark matte surface and imaged in blocks of 100 grains at a time. Acquisition was performed under controlled lighting conditions to achieve spectral stability and comparability between measurements. Finally, the data, in *.bil format, were stored on Network Attached Storage (NAS) devices for subsequent analysis.

Routines and functions were implemented to visualize the information contained in the hypercubes; the intensity image at 1475 nm was used—due to the contrast difference—to segment the rice grains. A square Region Of Interest (ROI) measuring 11 pixels per side was generated based on the center of gravity of each grain, and the mean spectral profile was extracted from it. Finally, the mean profiles for each grain, coded by variety, were stored in an Excel spreadsheet. The 500 rows of each variety in this matrix are taken to be its five images of 100 grains, in acquisition order; the image defines the unit of the grouped validation (Section 2.4.1).

2.2.2. Pre-treatment

Raw hyperspectral data often contain noise due to instrumental and environmental effects [29]; therefore, the following preprocessing steps were applied to each profile during extraction, before any of the analyses below.

- **Smoothing.** For noise reduction, the Savitzky-Golay filter was applied; this filter preserves the shape of the peaks and reduces high-frequency noise; see Equation 1.

$$y_{i,j} = \sum_{k=-m}^{m} c_k \, x_{i,\, j+k} \tag{1}$$

where $y_{i,j}$ is the smoothed value of the sample i at the wavelength index j; $x_{i,\,j+k}$ is the original value of the signal; c_k are the SG convolution coefficients determined by the degree of the polynomial p and the order of the derivative d; $m = (w-1)/2$ is the half-width of the window; and w is the total length of the window (an odd number of points), which corresponds to $w = 2m + 1$.

- **Normalization.** The Z-score normalization method was applied to the profiles, scaling the data points so that they have a mean (μ) of 0 and a standard deviation (σ) of 1, see Equation 2.

$$X_{Norm} = \frac{X - \mu}{\sigma} \tag{2}$$

where X_{Norm} is the normalized profile with Z-score, X is the raw profile, μ mean, and σ the standard deviation of the profile.

2.3. Exploratory analysis

As in Amigo et al. [30], Zhang et al. [31], Ali et al. [32] principal component analysis was performed on the column-centred matrix of pre-treated profiles (singular value decomposition in NumPy). The projection of the observations onto the first two principal components was visualized through a scatter plot, using distinct colors and markers per class, with confidence ellipses superimposed to assess dispersion and overlap between groups.

2.4. Model building and validation

The modeling step was designed to systematically compare different supervised classification strategies, identify the most informative spectral variables, and optimize previously implemented models using relevant variables. The pipeline comprised three complementary stages:

i Construction and training of the classification models,

ii Selection of relevant variables using three alternative feature-selection approaches

iii Evaluation of classification performance through averaged confusion matrices and quantitative metrics per-class.

For each step, the functions shared the same predictor matrix X, class vector Y, and spectral band vector Z, ensuring full reproducibility and direct comparability of the results across configurations.

2.4.1. Building

The classifiers were built and validated by five-fold cross-validation repeated six times (30 splits). Each split tests 20% of the grains of each variety and every grain is tested six times. Two versions of the division were compared. In the *random* version, individual grains were assigned to folds at random within each variety, as is usual in the field. In the *grouped* version, whole images of 100 grains were assigned to folds, so that each split holds out one image per variety and grains imaged together never appear on both sides. The random version estimates accuracy on grains from images already seen during training; the grouped version estimates accuracy on grains from another image, which is closer to how an instrument would be used.

a Linear Discriminant Analysis (LDA). This model, see Figure 2a, seeks a linear combination of variables that provides discrimination between classes [13]. It was fitted as a single multiclass model with a pooled covariance matrix (scikit-learn `LinearDiscriminantAnalysis`).

b Linear SVM on unstandardised bands (L-SVM): one-vs-one binary linear SVMs with hinge loss and $C = 1$ (scikit-learn `LinearSVC` in a `OneVsOneClassifier`), fitted to convergence. It differs from the SVM below mainly in the absence of standardisation.

c Support Vector Machine (SVM). This model constructs an optimal separation hyperplane between classes and the decision boundary which is determined by maximizing the margin between the hyperplane and the closest samples, known as support vectors (Figure 2b). In addition, kernel functions can be used to address nonlinear classification problems [15, 33]. A multiclass SVM with a linear kernel ($C = 1$, one-vs-one) was fitted on standardized bands (scikit-learn `SVC` after `StandardScaler`).

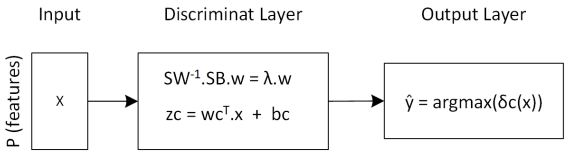
d Artificial Neural Network (ANN). In this case, interconnected processing units, called neurons, are organized into layers and learn nonlinear relationships between input variables and outputs by adjusting connection weights. In the specific case of the neural network of backpropagation, learning involves forward propagation of information followed by backward propagation of the prediction error to update the network parameters (Figure 2c) [20, 33]. A feed-forward network with standardized inputs and a single ReLU hidden layer of round($\sqrt{p}$) neurons, p being the number of bands (16 for the full spectrum, 3 for nine bands), was trained with L-BFGS (scikit-learn `MLPClassifier`).

2.4.2. Selection of relevant variables

Three alternative feature selection strategies were implemented to identify the most relevant spectral bands and to compare their effect on classification performance [13, 34]:

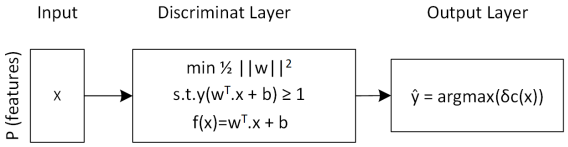
- Feature Selection by Neighborhood Component Analysis (FSCNCA). This supervised method assigns a weight to each feature based on its contribution to prediction. Features with larger weights are considered to be more relevant, whereas features with small weights can be removed to obtain a reduced representation of the original feature space. Feature-weighted NCA [35] was implemented in Python: a weight per band, distance $D_{ij} = \sum_r w_r^2 |x_{ir} - x_{jr}|$, kernel width $\sigma = 1$, regularisation $\lambda = 1/n$, weights initialised at 1 and optimised by gradient ascent (Adam, learning rate 0.02, 150 iterations) on 100 standardised profiles per variety. The bands were ranked by their weight and the k largest retained.

- Sequential Forward Selection (SFS). This is a wrapper-based method that starts with an empty feature subset and sequentially adds the feature that produces the greatest improvement according to a predefined model-performance criterion. The process progressively constructs a reduced feature subset without exhaustively evaluating all possible combinations. The implementation uses an LDA classifier as the evaluation criterion, starting from an empty feature set; iteratively evaluating all remaining candidate features and selecting, at each step, the one that maximizes the mean 5-fold cross-validated accuracy; the inner folds are formed from blocks of 10 consecutive grains, so that neighbouring grains are not split between fitting and scoring. The first k bands added are retained.



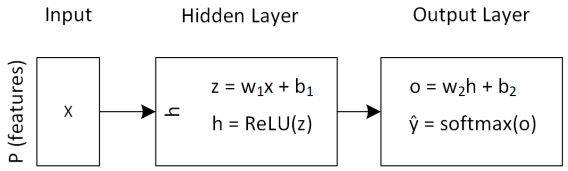
Where:
p = Number of predictor
C = Number of classes
SB = Between-class scatter matrix
SW = Within-class scatter matrix
wc = Discriminant coefficient of class c (c = 1, ... , C)
b = Bias term
δc(x) = Discriminant function of class c
ŷ = Predicted class label

(a)



Where:
p = Number of predictor
C = Number of classes
M = Number of binary classifiers C(C-1)/2
w = Weigth vector (normal to hyperplane)
b = Bias term
y = Binary class label ∈ {-1, +1}
SW = Within-class scatter matrix
f(x) = Decision function of the binary SVM
δc(x) = Agregated discriminant score for class c
ŷ = Predicted class label

(b)



where:
p = Number of predictor
h = Number of neurons (√p)
C = Number of classes
w_i, b_i = weights and biases of layer i
ŷ = Predicted class label

(c)

Figure 2: Schematic diagram for models a) LDA, b) SVM, and c) ANN

- Covering Array Feature Selection (CAFS). It is a supervised wrapper approach based on covering arrays, combinatorial structures characterized by high interaction coverage with a reduced number of configurations. CAFS uses the subset of experiments defined by the covering array to explore the feature space and identify the features that maximize a predictive-performance criterion, such as the F1 score. This enables the exploration of relevant feature interactions with substantially fewer combinations than an exhaustive search. The implementation is based on previously reported work by Castro et al. [34], and uses a Gaussian naive Bayes classifier scored by macro-F1 in an inner 5-fold cross-validation (folds formed from blocks of 10 consecutive grains, as for SFS). It was ported line by line to Python from the MATLAB code of Castro et al. [34] (binary covering arrays of strength 2; our generator reproduces theirs exactly). At each iteration the current band set is split according to the rows of the covering array, the best-scoring row becomes the new set, and the best subset over all iterations is returned; CAFS therefore sets its own subset size. The algorithm was configured with the following options:

  – Patience parameter (Patience = 3)
  – Minimum improvement threshold (MinImprovement = 1e-4)
  – Minimum number of features (MinFeatures = 10)
  – Maximum number of iterations (MaxIterations = 50)
  – Target number of features (TargetFeatures = 20)
  – Compactness penalty (CompactnessPenalty = 0.10)
  – ExcessPenaltyPower = 2
  – AdaptivePenaltyAlpha = 2

  Spectral preprocessing was also applied using standard normal variate (SNV) to reduce multiplicative scattering and baseline differences between samples, and a target-excess compactness penalty was included that discourages the selection of subsets larger than the target number of wavelengths.

Two uses of selection. Each method was applied in two ways. (1) *Once on all spectra*: SFS and FSCNCA retaining $k = 9$ bands, and CAFS with its own subset size, fitted on all 6000 profiles, as is common in the field. The test grains then take part in choosing the bands, which favours these subsets [27]; they are evaluated to show how large this advantage is. (2) *Within training*: each method fitted on the training part of a split only, so that the test grains take no part in choosing the bands; SFS and FSCNCA at $k \in \{5, 10, 20, 40\}$ and CAFS with its own size, on ten splits (two repetitions of the grouped five-fold cross-validation, drawn independently of the 30 splits of Table 2). **Random-subset baseline.** Because neighbouring NIR bands are strongly correlated, a small set of bands drawn at random may already carry most of the discriminant information. To judge whether a selection method identifies informative bands, rather than merely reducing their number, every

selected subset was compared on the same split, with one image per variety held out, against 30 subsets of the same size drawn uniformly at random from the 263 bands.

2.4.3. Performance evaluation

The classification performance was evaluated through two complementary procedures:

- Generation of average confusion matrices. For each repetition a confusion matrix is computed, and the resulting matrices are pooled across the N iterations to obtain a representative confusion matrix. Each row of this averaged matrix is then normalized to percentages, yielding a row-wise accuracy representation that facilitates the interpretation of per-class performance independently of class size and displayed as a heatmap.

- Calculation of quantitative per-class metrics. For each iteration, the following quantities were derived from the confusion matrix: true positives (TP, diagonal elements), false positives (FP, column sums minus TP), false negatives (FN, row sums minus TP), and true negatives (TN, total minus (FP + FN + TP)). From these, three standard metrics were calculated per class and per iteration: precision, defined as TP / (TP + FP); recall (sensitivity), defined as TP / (TP + FN); and the F1-score, defined as the harmonic mean 2·(Precision·Recall) / (Precision + Recall). The structure of the confusion matrix and these ratios are shown in Figure 3, and the metrics were calculated in a way similar to Castro et al. [33], see Equations 3 to 6; Equation 3 is the overall accuracy over all classes, and Equations 4–6 are per class.

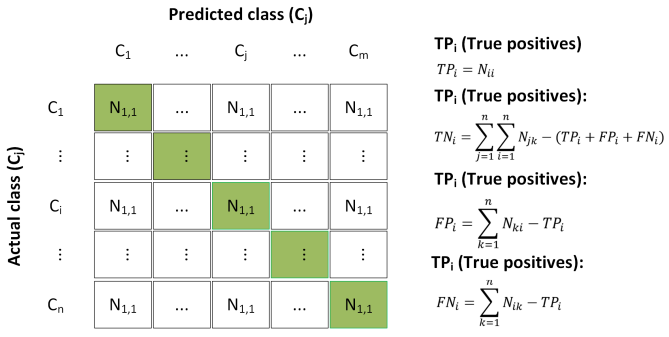


Figure 3: CM structure and main ratios

$$Accuracy\,(acc) = \frac{\sum_{i=1}^{K} TP_i}{\sum_{i=1}^{K}\sum_{j=1}^{K} N_{ij}} \tag{3}$$

$$Precision\,(pre) = \frac{TP_i}{TP_i + FP_i} \tag{4}$$

$$Recall\,(rec) = \frac{TP_i}{TP_i + FN_i} \tag{5}$$

$$F\text{-}measure_i\,(FM) = 2\left(\frac{pre_i \cdot rec_i}{pre_i + rec_i}\right) \tag{6}$$

Overall accuracy was defined as the proportion of test grains assigned to their correct variety, $\sum_c \mathrm{TP}_c/N$, and summarised together with the macro-averaged F1-score. The per-class one-vs-rest accuracy $(\mathrm{TP} + \mathrm{TN})/N$ was not used as a summary: with K classes and overall error e, its mean over classes equals $1 - 2e/K$ exactly, whatever the class balance, because every misclassified grain counts once as a false negative and once as a false positive. It therefore carries no information beyond overall accuracy, compresses differences by a factor $K/2$, and cannot fall below $1 - 2/K$: for $K = 12$ an overall accuracy of 97% appears as 99.5%, random guessing scores 84.7%, and a classifier that is always wrong still scores 83.3%. For each metric, the mean and standard deviation over the 30 splits were calculated; this procedure was repeated for each classifier and each feature configuration considered in the study (full profile and the three optimized subsets obtained in Section 2.4.2), allowing a systematic comparison of predictive performance between models and feature subsets and providing the empirical basis for selecting the most suitable combination of classifier and feature representation. We applied an explicit criterion to our analysis (Sections 3.4 and 3.5). The 30 splits reuse the same 60 images, so they are not independent observations and tests on them overstate the evidence [36]. Classifiers were therefore compared at the level of images: the accuracy of each image was averaged over the repetitions in which it was tested, the classifiers were compared in pairs over the 60 images with the Wilcoxon signed-rank test, and 95% confidence intervals for the differences were obtained by resampling images (10 000 bootstrap samples); random and grouped division were compared in the same way, paired by image. Band subsets were compared with random subsets on each split (paired t-test and Wilcoxon test over the splits); these comparisons share the same limitation, so their p-values indicate how consistent a difference is across splits rather than giving formal evidence. P-values were adjusted with the Holm procedure within each family (12 classifier comparisons; 4 drops from random to grouped division; 12 comparisons of the subsets selected once on all spectra with random subsets; 30 comparisons of training-only selections with random subsets), and a difference is reported as significant at an adjusted $p < 0.05$.

3. Results

3.1. Mean profiles

Figure 4 shows the mean profile of each variety after Savitzky-Golay smoothing and Z-score normalisation. The profiles are reflectance, so absorption bands appear as minima and as the falling edges that lead into them, while maxima are windows of comparatively weak absorption between bands. All twelve profiles share the same shape: a rising edge from 1000 nm to a reflectance maximum near 1120–1125 nm, a sharp fall to a minimum near 1195–1200 nm, a secondary maximum near 1300–1330 nm, a steep fall between about 1340 and

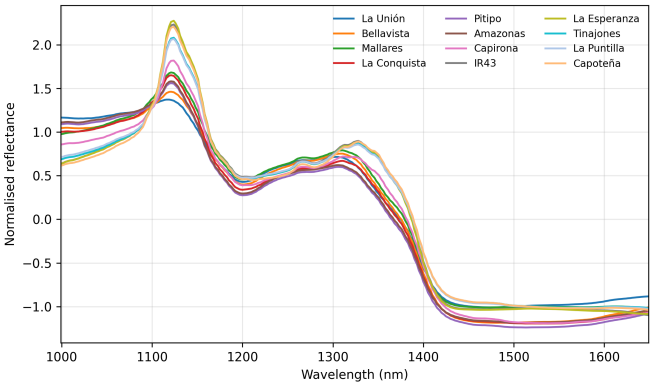


Figure 4: Mean profiles of each variety

1430 nm, and a low, nearly flat plateau from 1430 nm to the end of the range near 1650 nm, where the varieties differ little. Varieties differ mainly in amplitude rather than in position: the highest maxima near 1120 and 1300 nm belong to the same two or three varieties throughout, and the lowest to a different, consistent subset. In this range the varietal signal is therefore carried by the depth of the absorption features and the height of the windows between them, not by shifts in wavelength. The minimum near 1195–1200 nm lies in the second-overtone region of C–H stretching of CH_2 and CH_3 groups [37]. On rice grain, John et al. [38], surveying 500 brown rice accessions, assign 1196 nm to "the C–H second overtone corresponding to aliphatic hydrocarbons", and Onmarkhong et al. [15] assign 1202 nm to the second overtone of the symmetric C–H stretching of CH_3 groups, associated with amylose. The falling edge that leads into this minimum, from the maximum near 1125 nm, covers the band that Shao et al. [39] found in second-derivative absorbance spectra of rice flour at 1131–1155 nm and assigned to C–H stretching of starch (they call it the first overtone; standard tables place C–H second overtones in this region [37]). The C–H absorption near 1200 nm thus combines starch and lipid, and the single band does not by itself separate the two. The fall between 1340 and 1430 nm and the low plateau beyond it are the first-overtone O–H absorption of water, which Shao et al. [39] place at 1418 nm in rice flour, together with C–H combination bands near 1360–1420 nm [37]; the plateau also spans the first overtone of N–H stretching of protein near 1500–1530 nm and the O–H stretching of starch near 1580 nm [15]. The maxima near 1120 and 1300 nm are not absorption bands: their height reflects how deep the neighbouring bands are and how strongly the grain scatters. Band positions shift with the matrix, so we rely on the rice studies [15, 38, 39] for specific positions and use the general tables only at the level of spectral regions. The broad characteristic between approximately 1250 and 1340 nm should be interpreted as an overlapping response of the rice matrix rather than being assigned to a single constituent. The pronounced decrease in reflectance around 1400–1450 nm corresponds to the first overtone of OH stretching and has been associated with the OH groups of water, starch and amylose in rice [15]. At longer wavelengths, the characteristic around 1580 nm has been associated with O-H stretching involving intermolecular hydrogen bonding in starch and glucose [15]. Therefore, the spectral differences between the twelve varieties are mainly consistent with variations in starch matrix, amylose-related composition, moisture, and molecular interactions.

3.2. *Exploratory analysis*

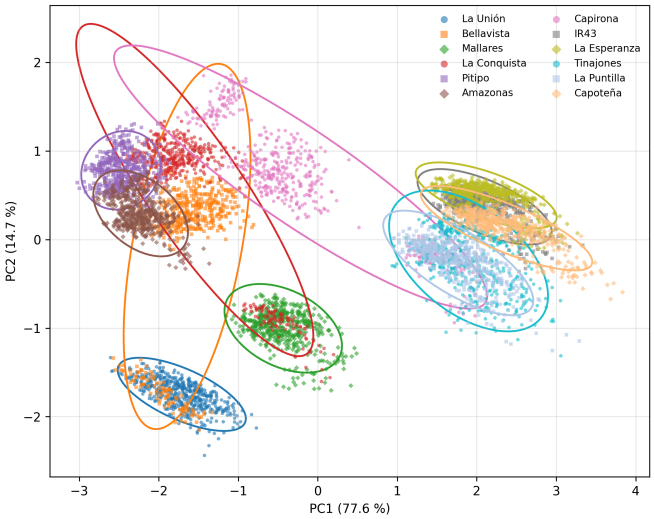


Figure 5: PCA for spectral profiles

Figure 5 shows the PCA score plot obtained from the NIR spectral profiles of the twelve rice varieties. PCA transforms highly correlated spectral variables into a reduced set of orthogonal components that capture the main sources of spectral variability. Consequently, samples with similar NIR profiles are projected close to each other, whereas differences in spectral shape and intensity lead to separation in the PCA space. This behavior has been demonstrated for rice using FT-NIR spectroscopy, where PCA revealed spectral patterns associated with varietal and geographical differences [4]. Thus, PCA summarizes the combined variations across the NIR spectrum rather than separating samples according to a single wavelength or chemical compound; similar studies using vibrational spectroscopy and PCA have shown that multivariate spectral fingerprints can reveal natural groupings and support subsequent classification or discrimination [3, 34].

Figure 5 projects the spectra onto the first two principal components, with a 95% confidence ellipse per variety. The first component separates two broad groups: seven varieties at negative scores (La Unión, Bellavista, Mallares, La Conquista, Pitipo, Amazonas, Capirona) and five at positive scores (IR43, La Esperanza, Tinajones, La Puntilla, Capoteña). Within the second group the second component partly separates IR43, La Esperanza and Capoteña (upper) from Tinajones and La Puntilla (lower), with heavy overlap inside each pair. Within the first group, La Unión forms a compact cloud at the bottom, Mallares a distinct cloud at the centre, and Pitipo and Amazonas overlap at the upper left. Three varieties – Bellavista, La Conquista and Capirona – each appear as two separate clouds

rather than one: part of Bellavista lies with La Unión and part near Amazonas, part of La Conquista near Pitipo and part with Mallares, and Capirona in two lobes at the upper centre. This splitting is what stretches their ellipses across the plot; these three varieties are the ones with an image classified almost entirely as another variety (Section 4.4). The first two principal components therefore group the twelve varieties into three or four blocks rather than resolving them individually, consistent with the classifiers' reliance on many discriminant dimensions (Section 2.4.1) rather than on a simple two-dimensional separation. In a principal component analysis of the mean-centred matrix (263 channels), the first component alone carries 77.6% of the total variance, the first two together 92.4%, five components are needed to reach 99.4%, and eleven for 99.9%. The limited class separation visible here is therefore a two-dimensional cut through a signal that needs far more than two components to be summarised adequately, which is consistent with the classifiers using many discriminant or SVM dimensions rather than two.

3.3. Feature selection

Figure 6 marks the bands selected by each method as vertical lines on the mean-profile background of Figure 4. FSCNCA: 1031, 1033, 1038, 1114, 1116, 1170, 1543, 1566 and 1641 nm. SFS: 1033, 1102, 1116, 1126, 1151, 1244, 1385, 1430 and 1649 nm. CAFS: 1050, 1068, 1138, 1141, 1425, 1435, 1626, 1644 and 1649 nm. All three methods place two to four bands on the rising edge and maximum between 1100 and 1150 nm, and one near the upper end of the range (1641–1649 nm). SFS and CAFS each place two bands in the fall into the O–H region (1385–1435 nm), where FSCNCA places none, choosing instead three bands near 1035 nm and two on the plateau at 1543–1566 nm. Only three wavelengths are shared exactly by two methods (1033 and 1116 nm by FSCNCA and SFS, 1649 nm by SFS and CAFS). That three reasonable methods agree so little on which bands matter, while reaching comparable accuracy (77–84% with the SVM with one image per variety held out, Table 3), is itself informative about how redundant the bands are; Section 3.5 tests this directly against randomly drawn subsets. This last region is particularly relevant because absorption near 1450 nm is associated with the first overtone of O–H stretching and contributions from water, starch, and hydroxyl groups of amylose [15]. The differences among the selected subsets reflect the distinct search strategies of the algorithms. FSCNCA and SFS tend to identify individual wavelengths according to their contribution to class discrimination, whereas CAFS explores combinations of wavelengths through covering arrays. Consequently, CAFS can retain variables from different spectral regions whose joint response is informative even when their individual contribution is limited. This behavior is consistent with the rationale for the selection of wavelengths based on covering-arrays reported by Castro et al. [34], where combinations of NIR wavebands were used to preserve discriminative compositional information. It is important to note that the selected wavelengths should not be interpreted as exclusive markers of individual compounds because the NIR bands are broad, overlapping, and highly correlated. Instead, their selection indicates spectral regions whose combined CH and OH responses capture differences in starch, amylose, moisture, and the molecular environment of the samples.

3.4. Classification

Table 2 gives the overall accuracy (Equation 3) of the four classifiers on the full spectrum. Under the random division the SVM reaches 97.2% and the ANN 97.1%, which correspond to 99.53% and 99.51% in per-class one-vs-rest accuracy, and LDA and the L-SVM reach 90.4% and 90.2%. With one image per variety held out, accuracy falls to 87.8% for the SVM, 88.2% for the ANN, 86.7% for LDA and 84.2% for the L-SVM, and the standard deviation across splits grows from 0.5–0.8 to 6–7 points. Compared over the 60 images (Section 2.4.3), the drop from random to grouped division is significant for all four classifiers (adjusted $p < 10^{-4}$): 9.4 points for the SVM (95% confidence interval 4.3–15.5), 9.0 for the ANN (4.2–15.0), 6.1 for the L-SVM (3.1–9.9) and 3.8 for LDA (1.7–6.2). Under the random division the SVM exceeds LDA and the L-SVM by 6.8 and 6.9 points (intervals 3.0–11.7 and 3.2–11.6; $p \leq 0.0003$); the lead of the ANN over them is as large (6.7–6.8 points) but less consistent across images ($p = 0.067$), the SVM and the ANN differ by only 0.1 points (consistently across images, $p = 0.011$, but negligibly), and LDA and the L-SVM do not differ ($p = 1.0$). With one image held out, the SVM, the ANN and LDA no longer differ (differences of 0.2–1.4 points, with intervals that include zero), and only the lead of the SVM over the L-SVM remains (3.7 points, interval 1.4–6.3, $p = 0.023$). The advantage of the SVM and the ANN under the random division therefore comes from the grains they share with training images.

Figure 7 shows the confusion matrices, pooled over the 30 splits. The errors concentrate in a few pairs of varieties: with the SVM and the ANN under the random division, IR43 and La Esperanza are taken for each other in 7–8% of cases, La Conquista for Mallares in 2–5%, Tinajones and La Puntilla in 3–5% and Pitipo and Amazonas in 1–2%, and LDA and the L-SVM confuse the same pairs at about twice these rates (IR43 and La Esperanza 14–18%). Other confusions involve whole images. Under the random division LDA and the L-SVM already assign about 20% of Bellavista to La Unión and 20% of La Conquista to Mallares – one image in each case – whereas the SVM and the ANN, trained on other grains of the same images, do so for only 1–5%; with the images held out, all four classifiers lose them, and Capirona is taken for La Puntilla in 14–20% of cases, again one image (Section 4.4). This is the mechanism behind most of the lead of the SVM and the ANN under the random division.

3.5. Random-subset baseline for band selection

Each selection method was compared against randomly drawn band subsets of the same size, with one image per variety held out rather than under the random division, since an advantage from selecting bands only matters if it survives on grains from a new image. Each method was fitted on the training part of each of 10 splits (two repetitions of the grouped five-fold cross-validation, drawn independently of the 30 splits of

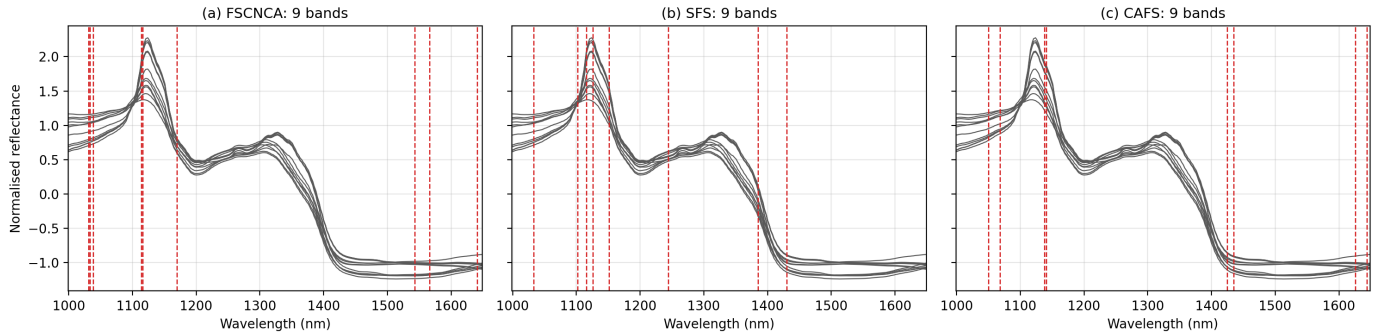


Figure 6: Relevant features using a) FSCNCA, b) SFS, c) CAFS: the nine bands selected by each method, fitted once on all 6000 profiles (red dashed lines), on the mean reflectance profiles of the twelve varieties.

Table 2: Overall accuracy (%, mean ± SD over 30 splits: six repetitions of a five-fold cross-validation) of the four classifiers on the full spectrum, with grains divided at random and with one image per variety held out. L-SVM: linear SVM on unstandardised bands (Section 2.4.1). Last column: the per-class one-vs-rest accuracy of the SVM, for comparison with studies that report it.

Division	SVM	L-SVM	LDA	ANN	SVM, OvR
Random	97.2 ± 0.5	90.2 ± 0.7	90.4 ± 0.8	97.1 ± 0.5	99.5 ± 0.1
One image held out	87.8 ± 6.3	84.2 ± 7.0	86.7 ± 6.7	88.2 ± 6.5	98.0 ± 1.1

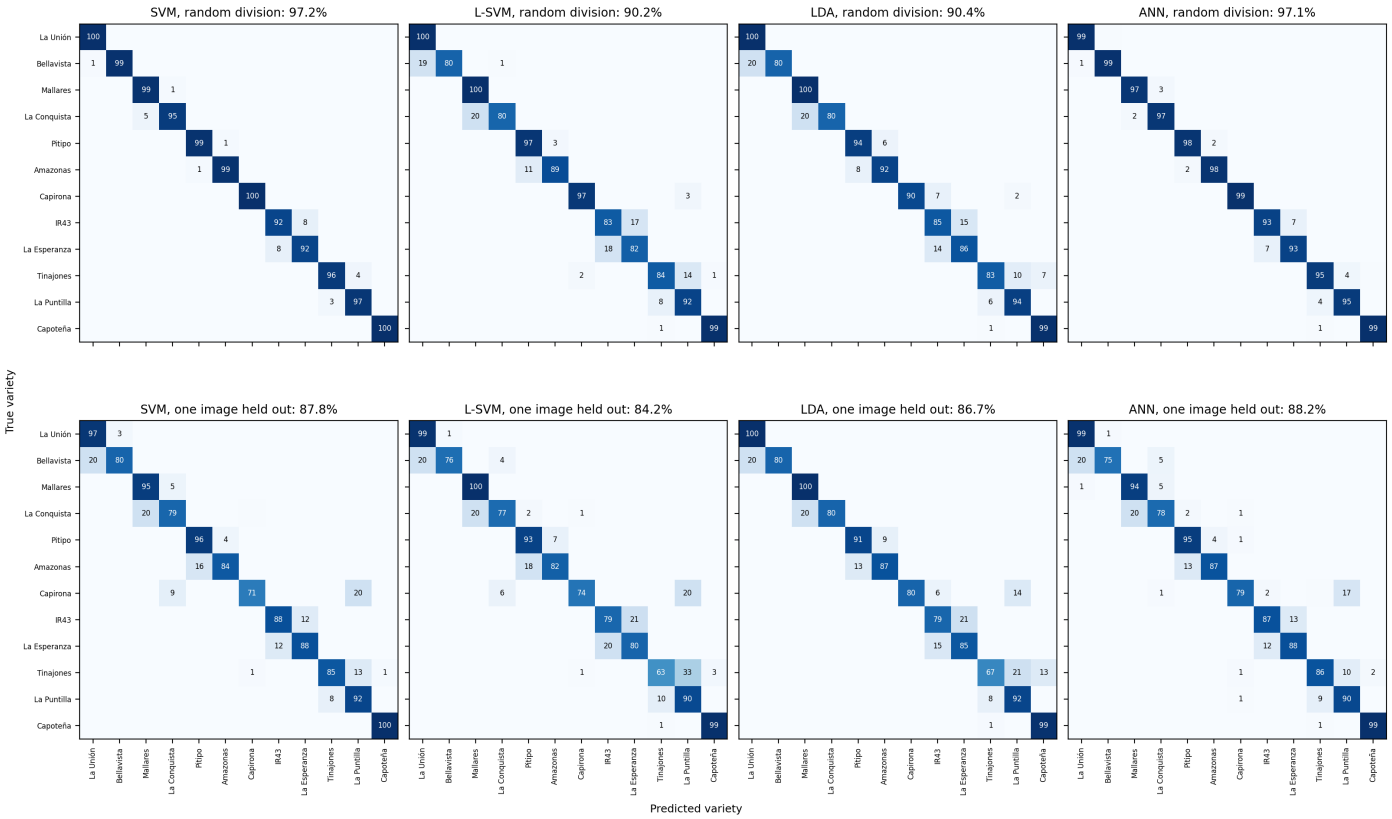


Figure 7: Confusion matrices (percent of the grains of each true variety, pooled over the 30 splits) of the four classifiers on the full spectrum: random division (top) and one image per variety held out (bottom). Panel titles give overall accuracy. Values below 0.5% are not shown.

Table 2), with any internal cross-validation grouped in blocks of 10 rows; the subset it returned was evaluated on the test part with the SVM and the LDA classifier (the ANN and the L-SVM were left out to limit the cost: 1200 fits per classifier and method), and compared against 30 randomly drawn subsets of the same size on the same split. Because the internal cross-validation groups blocks of 10 rows rather than whole images, grains of one image can still sit on both sides of the internal criterion, which may reward bands specific to an acquisition. The three methods faced the same splits, and SFS and FSCNCA the same random subsets. SFS and FSCNCA were evaluated at $k \in \{5, 10, 20, 40\}$ bands. CAFS (Section 2.4.2; a covering array of 12 rows for 263 bands in the first iteration) sets its own subset size: the set shrinks from 263 to about 137, 80, 39, 24, 16 and 14 bands, and the algorithm returns the best-scoring subset over all iterations (7–25 bands). The subset of each iteration and the returned subset were compared against random subsets of the same size.

Tables 4 and 5 and Figure 8 summarise the results. SFS beats the random mean at every k, by 6–9 points at $k = 5$ and by at most 0.6 points at $k = 40$. FSCNCA is never clearly better than random: 1.5–2.6 points ahead at $k \leq 10$, and level with or up to 1.4 points below random at $k = 20$ and 40. CAFS falls 1.0–3.0 points below the random mean at every iteration shown, and the subset it returns – 19 bands on average, from 7 to 25 depending on the split – is also below random (80.1% against 82.4% with the SVM, 79.2% against 82.2% with LDA). In every case the selected subset falls inside the 5–95% range of the random subsets. The advantage of selecting bands over drawing them at random is therefore at most modest, largest for SFS with very few bands, and vanishes as more bands are kept. Paired over the splits with Holm adjustment across the 30 comparisons, the paired t-test finds SFS better than random at 5 bands (by 6.2 points with the SVM and 8.8 with LDA; adjusted $p = 0.002$ for both), at 10 bands (3.6 and 6.0 points; $p = 0.007$ and 0.0007) and at 20 bands with LDA only (2.3 points; $p = 0.0005$); FSCNCA never better, and worse at 40 bands with the SVM (1.4 points, $p = 0.010$); and CAFS never better, but worse than random at iterations 1–4 with the SVM (by 1.6–2.2 points, adjusted $p \leq 0.02$) and at iterations 2 and 3 with LDA (2.4 and 2.7 points, adjusted $p = 0.042$ and 0.006); its returned subset does not differ from random ($p = 0.18$ with the SVM, 0.56 with LDA). With 10 splits the Wilcoxon signed-rank test cannot reach an adjusted $p < 0.05$ for any of the 30 comparisons (its smallest attainable unadjusted p is 0.002), so these differences rest on the t-test alone, and the splits share images, so they are not independent. The nine-band subsets obtained by applying each method once to all 6000 spectra (Figure 6) were tested directly against random nine-band subsets with one image per variety held out (Table 3). The SFS subset beats the random mean with the three linear classifiers, by 3.6–7.3 points, and outperforms 86–99% of the random subsets, but not with the ANN (+0.1 points), whose hidden layer has only three units with nine bands. The FSCNCA subset is ahead of the random mean by 1.3–3.8 points with the linear classifiers and 7.1 points behind it with the ANN. The CAFS subset is ahead with the L-SVM (3.2 points) but behind with the SVM and LDA (3.3 and 2.4 points). All differences except those of SFS and CAFS with the ANN are consistent across splits (Holm-adjusted $p \leq 0.004$). These subsets were selected with all the spectra, so part of their advantage may come from the test grains taking part in the selection: when selection is fitted on the training part of each split, as above, FSCNCA no longer beats random subsets and SFS does so mainly with 5–10 bands.

4. Discussion

4.1. Validation design and metric

The accuracies obtained under random and grouped division estimate different quantities. Under the random division the test grains come from images also represented in training; with an image held out they come from another image of the same lot, which is closer to how an instrument would be used, though still optimistic for a new lot, season or session. The gap between them, about 9 points for the SVM, depends on how much grains imaged together share, a property of the acquisition rather than of the classifier, so it cannot be carried to another data set as a correction. The same distinction bears on the choice of classifier: the 7-point lead of the SVM and the ANN over LDA and the L-SVM under the random division disappears against LDA with an image held out, where the SVM, the ANN and LDA no longer differ, and shrinks to 3.7 points against the L-SVM. The per-class one-vs-rest accuracy, in turn, compresses differences by a factor of six for twelve classes and cannot fall below 83.3%. The studies in Table 1 report accuracies of 90–100%; where a study does not state whether grains from one image or session were kept together, or which accuracy it reports, its figures and ours cannot be compared as they stand.

4.2. Band selection

The three selection methods retained different bands (Figure 6). Nine-band subsets selected once with all the spectra beat random nine-band subsets clearly only for SFS with the linear classifiers; FSCNCA led by a few points at most and CAFS was above or below random depending on the classifier (Table 3); because they were chosen with all the spectra, the training-only analysis is the fairer test. Fitted on training data only and tested on held-out images, SFS beat randomly drawn subsets of the same size at 5 and 10 bands, by 4–9 points – the range that matters for an instrument with few channels – and at 20 bands with LDA, but not at 40; FSCNCA led random subsets by at most 2.6 points and not at all from twenty bands on; and CAFS trailed them by 1.0–3.0 points at every size. Neighbouring NIR bands are highly correlated, so a well-spread subset of adequate size retains most of what a ranked subset does. A band-selection result on spectra of this kind should therefore be reported against a random-subset control of the same size.

4.3. Chemical interpretation

The features that differ most between the mean profiles are the C–H second-overtone absorption near 1200 nm, shared by starch and lipid, and the O–H absorption of water above about 1400 nm (Section 3.1). These differences are consistent with

Table 3: Overall accuracy (%, mean ± SD over the 30 splits, one image per variety held out) of the nine-band subsets obtained by applying each selection method once to all the spectra (Figure 6), against 30 random nine-band subsets per split (mean, and 5–95% range over all random draws). Fraction: share of random subsets outperformed, averaged over splits. The subsets were selected with all the spectra, which favours them. With nine bands the ANN has three hidden units.

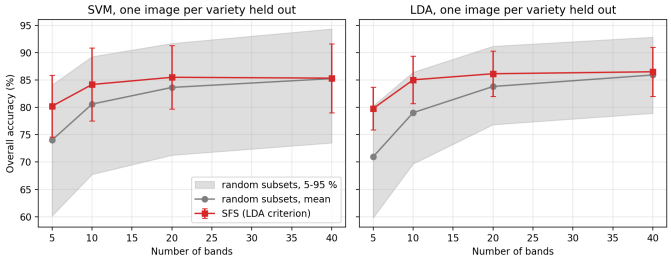
Classifier	Subset	Accuracy	Random mean (5–95%)	Difference	Fraction
SVM	FSCNCA	81.9 ± 7.7	80.6 (68.7–91.3)	+1.3	0.61
	SFS	84.2 ± 7.2	80.6 (68.7–91.3)	+3.6	0.86
	CAFS	77.3 ± 6.7	80.6 (68.7–91.3)	−3.3	0.16
L-SVM	FSCNCA	72.8 ± 7.8	69.0 (55.4–80.0)	+3.8	0.77
	SFS	76.2 ± 7.1	69.0 (55.4–80.0)	+7.3	0.99
	CAFS	72.1 ± 6.9	69.0 (55.4–80.0)	+3.2	0.72
LDA	FSCNCA	80.2 ± 7.5	78.5 (64.7–89.8)	+1.8	0.67
	SFS	85.1 ± 7.3	78.5 (64.7–89.8)	+6.6	0.98
	CAFS	76.1 ± 7.1	78.5 (64.7–89.8)	−2.4	0.28
ANN	FSCNCA	64.2 ± 7.4	71.3 (56.4–84.4)	−7.1	0.23
	SFS	71.4 ± 8.8	71.3 (56.4–84.4)	+0.1	0.49
	CAFS	72.1 ± 7.0	71.3 (56.4–84.4)	+0.8	0.51

Table 4: Overall accuracy (%, mean ± SD) of SFS- and FSCNCA-selected bands against 30 randomly drawn subsets of the same size, with one image per variety held out (10 splits; same splits and random subsets for both methods). The SD of the random subsets is over 300 evaluations (30 subsets in each of 10 splits), that of SFS and FSCNCA over the 10 splits.

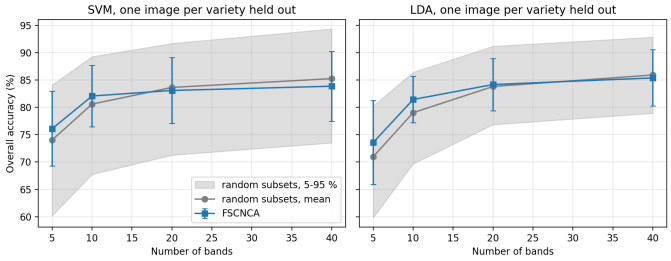
Bands k	SVM			LDA		
	SFS	FSCNCA	Random	SFS	FSCNCA	Random
5	80.2 ± 5.6	76.1 ± 6.8	74.0 ± 7.4	79.8 ± 3.9	73.6 ± 7.7	71.0 ± 6.3
10	84.2 ± 6.7	82.1 ± 5.6	80.6 ± 6.7	85.0 ± 4.4	81.4 ± 4.2	79.0 ± 5.2
20	85.5 ± 5.8	83.1 ± 6.0	83.6 ± 6.2	86.2 ± 4.2	84.2 ± 4.8	83.8 ± 4.5
40	85.3 ± 6.3	83.9 ± 6.4	85.3 ± 6.1	86.5 ± 4.5	85.4 ± 5.2	85.9 ± 4.8

Table 5: Overall accuracy (%, mean ± SD over splits) of CAFS-selected bands against 30 randomly drawn subsets of the same size, with one image per variety held out (10 splits). CAFS (settings of Section 2.4.2) sets its own subset size, so sizes are means over the splits that reached each iteration; iterations 7–9, reached by 5, 2 and 1 of the 10 splits, are not shown. Last row: the subset CAFS returns, the best-scoring over all iterations.

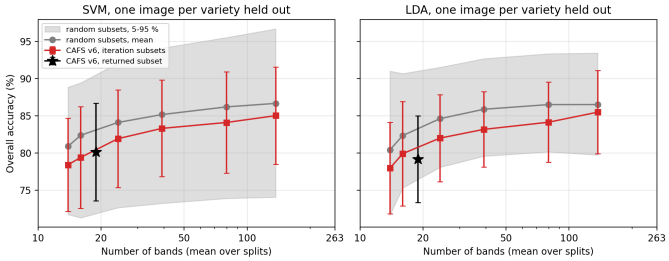
Bands (mean)	Splits	SVM		LDA	
		CAFS	Random	CAFS	Random
137	10	85.0 ± 6.5	86.7 ± 6.2	85.5 ± 5.6	86.5 ± 4.9
80	10	84.1 ± 6.8	86.2 ± 6.1	84.1 ± 5.4	86.5 ± 4.8
39	10	83.3 ± 6.5	85.2 ± 6.1	83.2 ± 5.1	85.9 ± 4.6
24	10	81.9 ± 6.5	84.1 ± 6.0	82.0 ± 5.8	84.6 ± 4.5
16	10	79.4 ± 6.8	82.4 ± 5.8	79.9 ± 7.0	82.4 ± 4.7
14	10	78.4 ± 6.3	80.9 ± 5.7	78.0 ± 6.1	80.4 ± 5.7
Returned, 19 (7–25)	10	80.1 ± 6.6	82.4 ± 5.7	79.2 ± 5.8	82.2 ± 5.2



(a) SFS



(b) FSCNCA



(c) CAFS

Figure 8: Overall accuracy of band subsets selected by (a) SFS, (b) FSCNCA and (c) CAFS (coloured), against 30 randomly drawn subsets of the same size (grey mean and 5–95% band), with one image per variety held out, for SVM (left) and LDA (right). In (c) the horizontal axis is logarithmic and gives the mean subset size per CAFS iteration (iterations reached by at least 9 of the 10 splits); the star is the subset CAFS returns.

differences in starch structure, lipid and moisture between varieties, but this paper does not test that attribution, which would require composition measured on the same grains that were imaged.

4.4. Limitations

Each variety is a single lot imaged in one session, so variety is confounded with lot and acquisition time. The confused pairs have consecutive codes, which suggests that the varieties were imaged in that order and that slow changes in the instrument contribute to the similarity of neighbouring varieties. Three of the 60 images – image 5 of Bellavista, image 1 of La Conquista and image 5 of Capirona – are classified almost entirely as La Unión, Mallares and La Puntilla when held out, which points to a labelling or sample slip that only the acquisition records can confirm; they are kept here with their labels as received. They are the extreme cases of a wider image effect: three more images (image 1 of Amazonas, image 1 of La Esperanza and image 3 of Tinajones) have fewer than two thirds of their grains classified correctly by every classifier. Finally, the grouping into images of 100 grains follows the acquisition protocol and has not been checked against image identifiers.

5. Conclusion

Twelve Peruvian rice varieties can be distinguished from NIR-HSI spectra of single grains, with an overall accuracy of about 97% for the linear SVM and the ANN when grains are divided at random and about 88% when whole images are held out; LDA reaches 90% and 87%, and the L-SVM 90% and 84%. Reporting the per-class one-vs-rest accuracy instead would show 99.5% and hide most of this difference. With images held out, the SVM, the ANN and LDA do not differ. Band selection reduced the number of bands without consistently identifying better ones: fitted on training data only, only SFS beat randomly drawn subsets of the same size, and only with few bands; nine-band subsets selected once with all the spectra did better for SFS, but the test grains took part in choosing them. Reporting overall accuracy, validating on images not used for training, and benchmarking band selection against random subsets would make accuracies comparable across studies.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used Claude Code (Anthropic, Claude Opus 5.5 model) to write and run the analysis code and statistics and to draft parts of the text. The authors reviewed and edited all content, verified every reported result and reference, and take full responsibility for the content of the published article. Code and a dated record of the analysis are available at `https://github.com/In-Silico-RG/Arroz_NIR_Replication`.

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